# PYTHON DICTIONARY

## ➤ WHAT IS DICTIONARY??

A dictionary in Python is a collection of key:value pairs. Curly brackets "{ }" are used for the construction of the dictionary. The key and its value are separated by the colon(:) and each pair is separated by the comma..

Example:

```
details = {'first_name':'Rohit', 'last_name':'Kumar', 'emp_id':200,
           'bill':200.2}
print(details)
```

Output:

```
{'first_name':'Rohit', 'last_name':'Kumar', 'emp_id':200, 'bill':200.2}
```

## ➤ ACCESS DICTIONARY ITEMS

- USING KEYS :⇒ To access dictionary items, we use the key inside the square brackets.

Example:

```
emp_details = {'first_name': 'Rahul', 'last_name': 'Kumar',
               'emp_id': 200}
print (emp_details ['emp_id'])
```

Output:

200

- USING get() Method :⇒ Using the get() method, we do not get any key error if the key does not exist.

Example:

```
emp_details = {'first_name': 'Rahul', 'last_name': 'Kumar', 'emp_id':
                                                              200}
print (emp_details.get ('first_name'))
print (emp_details.get ('salary'))
```

Output:

Rahul
None

- The items() Method :⇒ The items() method returns the view of tuples, and each tuple contains the key and its value.

Example:

```
numbers = {'1': 'One', '2': 'Two', '3': 'Three'}
print (numbers.items())
```

Output:

```
dict_items ([('1', 'One'), ('2': 'Two'), ('3': 'Three')])
```

- The keys() Method :⇒ The keys() method returns the view of keys.

Example:

```
numbers = {'1':'One', '2':'Two', '3':'Three'}
print(numbers.keys())
```

Output:

```
dict_keys(['1', '2', '3'])
```

- The values() Method :→ The values() method returns the view of values in the dictionary which can be traversed using the for loop.

Example:

```
numbers = {'1':'One', '2':'Two', '3':'Three'}
print(numbers.values())
```

Output:

```
dict_values(['One', 'Two', 'Three'])
```

- for loop :→ We can use the for loop to loop through the items of the dictionary.

Example:

```
numbers = {'1':'One', '2':'Two', '3':'Three'}
for i in numbers
    print(i)
```

Output:

1
2
3

We can use the items(), keys() or values() method if we want items, keys or values in output.

## ➤ DICTIONARIES ARE ORDERED

Dictionaries are ordered from Python version 3.7, and before that dictionaries used to be unordered.

## ➤ CHANGEABLE

Dictionaries are changeable, means we can change, add or remove items in the dictionary any time.

## ➤ DICTIONARY KEYS

Dictionary keys are special because of these properties:-

(1) UNIQUE : The key of the dictionary can not be repeated.

Example:

```
numbers = {'1':'One', '2':'Two', '1':'Three'}
print(numbers)
```

Output:

```
{'1':'Three', '2':'Two'}
```

(2) ONLY MUTABLE ELEMENTS : The key in the dictionary can have only immutable elements like number, string, tuple, etc. Mutable elements like a list or the dictionary itself are not allowed as the key.

Example:

```
numbers = {["Three","Four"]:["Three","Four"]}
print(numbers)
```

Output:
TypeError: unhashable type: 'list'

## ➤ MEMBERSHIP TEST :⇒

Membership operators in and not in are used to check whether a key is present in the dictionary or not.

```
numbers = {1: "One", 2: "Two", 3: "Three"}
print (1 in numbers)
print (2 not in numbers)
```

Output:
True
False

## ➤ NESTED DICTIONARY

A nested dictionary means when we have a dictionary inside another dictionary.

Example:

```
employees = {
    "Rohit": {
        "age": 55
        "gender": "Male"
    }
    "Steve": {
        "age": 57,
        "gender": Male"
    }
}

print (employees["Steve"])
```

```
print (employees["Rohit"]["age"])
```

Output:

```
'age':57, 'gender':'Male'}
55
```